

3-02-26 Network Layer :-

→ Logical Addressing :

ip address , Source / Destination

Unique address Eg: 192.168.111.1

→ Packet Formation :

Encapsulate whole data into packet
source address , Destination address

→ Routing :

Algorithm uses for ^{shortest, fastest} best path

Router used for pass packets

→ Packet Forwarding :

One router passes one
packet to another router.

→ Fragmentation & Reassembly :

When packet size is larger
than network capacity ^{is} the
fragmentation process.

In receiver side the frag-
mented are ~~ab~~orderly
received is called reassembly.

→ Internetworking :

The data are divided
into multiple network.

→ Connectionless device :



No guarantee for packet delivery, order etc.

Transport Layer :-

→ Heart of OSI Model

→ Segmentation & Reassembly :

Segmentation is process of converts data into segments. it is labeled by number.

Reassembly is possible by arrange the segments using number.

→ End to End delivery :

From one device to another device is called end to end delivery.

Port Number is used for ^{understand which} knowing app. is running

→ Error Control

→ Flow Control

→ Connection Control

→ Connection oriented
the connection is established before data transfer.

→ Connectionless

the just sends data, no acknowledgement recieved.

9-2-26 Session Layer :-

- Main aim is establish, maintain, synchronize b/w communicating system
- Manage Synchronize
- Streams of data are marked and resynchronized properly.

Functions :

Dialog Control :- Communication should be either half duplex or full duplex.

Token Management :- Prevent two parties from attempting the same critical operation at the same time.

A special control signal called a token is used.

Only the device hold the token is allowed to:

- Modify data
- Perform critical operations
- Other devices must wait until the token is released.

Synchronization : Allows a process to add checkpoints which are considered as synchronization points into stream of data.

Presentation Layer :-

- Data is transmitted in such a way that the receiver can understand and correctly use the information.
- It handles the representation, format meaning of data b/w com. sys.
- Different system may use different languages, character sets, and data representation.

Need for PL :-

Communication system may use diff.

It ensures platform-independent communication.



Functions :

- Translation : No. of characters → bits
- Encryption & Decryption :
- Compression : Reduce bandwidth require for transmission.
Reduce no. of bits
Improve transmission speed.

Application Layer :-

Functions :

- Mail Service
- Network Virtual Terminal : Remotely we can access host.
- Directory Services : Access for global information.
- File Transfer, Access and Management (FTAM)



10-2-26 Quality of Service (QoS)

- It is a techniques used in computer networks to control and management network traffic so that important applications to get better performance than less important ones.
- QoS ensures, reliable, predictable and efficient data delivery.

Why QoS uses (important)

In different network different data are transformed QoS select which should be transformed.

Quality of Service Parameters :-

Bandwidth : Maximum Data rate

Delay : time that taken to travel from source to destination

Jitter : Variation in delay

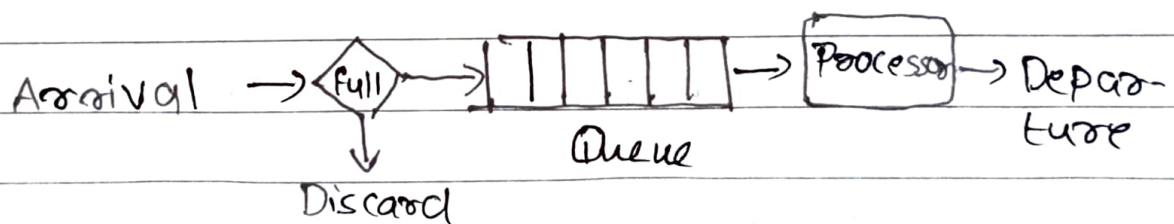
Packet Loss : ~~to~~ Data loss during transmission.

Techniques to Improve QoS :-

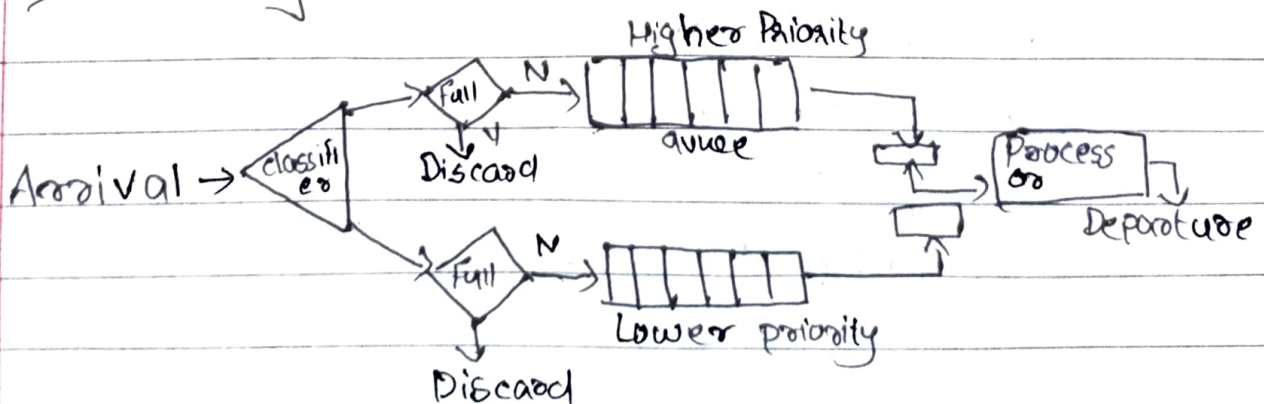
i) Scheduling

If more packets moves to ^{needs to} moves to

a) FIFO Queuing



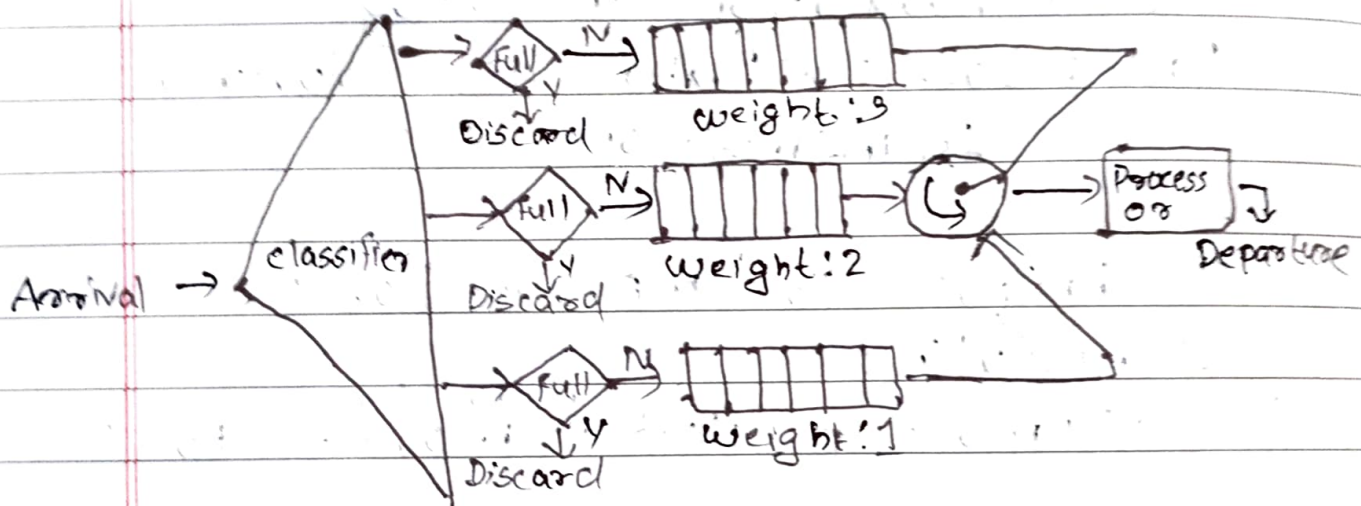
b) Priority Queuing



starvation occurs

c) Weighted Fair Queuing

→ Avoid Starvation



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ii) Traffic Shaping

It is a mechanism to control the amount and the rate of the traffic sent to the network.

Two techniques

- Leak bucket

- Token bucket

a) Leaky bucket Algorithm

I/P - worst

O/P - Constant

Why needed?

- Users send data in bursts.
- Burst traffic issues:

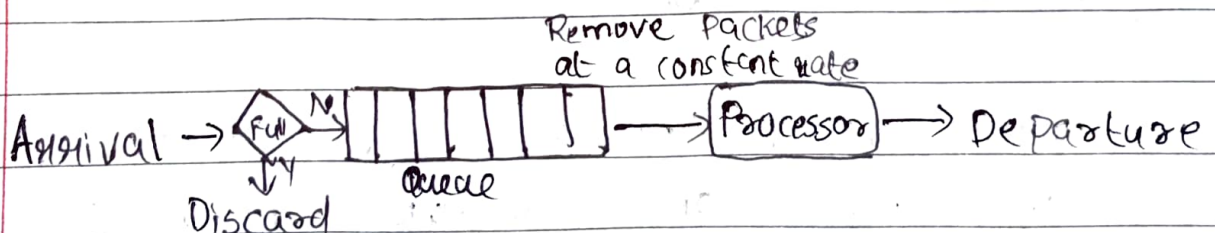
Congestion

Packet loss

High delay

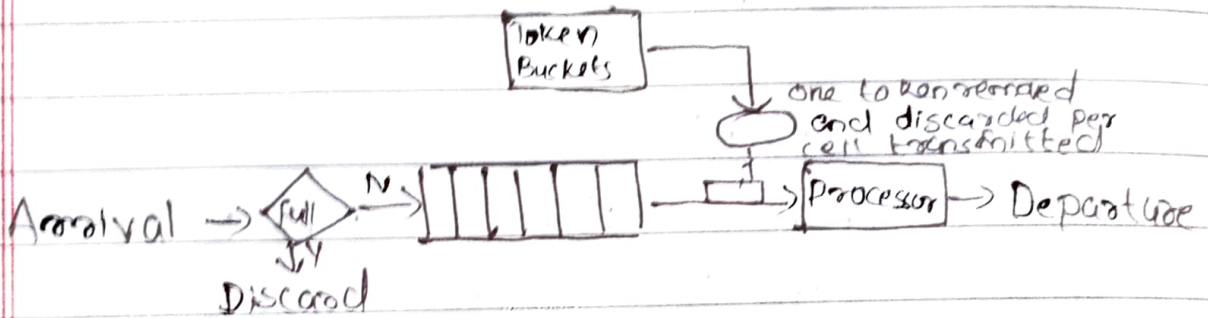
Jitter

- Convert bursty input to smooth output.
- Enforce committed bandwidth
- Protect the network.



b) Token Bucket Algorithm

To control the amount of data transfer.



iii) Resource Reservation

A data flow needs network resource such as:

Buffer space

Bandwidth

CPU time

Queue space

iv) Admission Control

used to accept or reject a new data flow based on available network resource. Router or switch used to check.

Router checks

Bandwidth, Buffer space, CPU processing power, Queue capacity.

If sufficient resource are available $\rightarrow$ Flow is accepted
if no $\rightarrow$ Flow is not accepted

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Application Layer Protocols

1) DNS - Domain Name System

Directory service that provides a mapping b/w the name of a host on the network and its numerical address.

It requires internet

Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots. It translates domain name to IP address.

Three ^{section} types of Domain name:

i) Generic Domain

Generic behavior

Eg: com - Commercial

edu - education

gov - government

info - Information Service Providers

ii) Country Domain

Used two character

It represents Country

Eg: In - India

iii) Inverse Domain

mapping an address to a name.

Working of DNS :-

Client sends request to server.

Server sends response to client.

Client ^{sends request} give a name which converted to IP address. Known as a Forward DNS lookup.

IP address which is converted into a name known as a reverse DNS lookup.

DNS implements a distributed database to store the name of all the hosts available on the internet.

Sender request contain host name then DNS resolver send a request to DNS server. If it is found it displays otherwise it checks for other server to get the IP address. Until IP address found the process continues . .